

# Evaluating Plasma p-tau217 as an Endpoint for Alzheimer Disease Clinical Trials

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## Abstract

### Background and Objectives

Plasma phosphorylated tau 217 (p-tau217) levels have been shown to parallel neurofibrillary tangle and cognitive decline over time. Although recent clinical trials have demonstrated exploratory drug effects on plasma p-tau217, little is known about plasma p-tau217 effect size and performance as an endpoint in Alzheimer disease (AD) clinical trials. Therefore, the objective of this study was to assess the longitudinal performance and potential utility of plasma p-tau217 as a primary endpoint in early-stage AD clinical trials.

### Method

This retrospective study included participants from 4 cohorts: ADNI (a multisite observational study), BICWALZS (a South Korean memory clinic-based cohort), MYHAT-NI (a Southwestern Pennsylvania population-based cohort), and WRAP (older adults at risk for AD). Eligible participants had plasma p-tau217 measurements at least at 2 timepoints, along with baseline A $\beta$  PET imaging and clinical assessments. Linear mixed-effects models were used to assess associations between plasma p-tau217 trajectories and clinical or biomarker outcomes. Effect size was defined as the mean annual rate of change in p-tau217 divided by its standard deviation. We calculated the sample size required for a hypothetical clinical trial designed to detect a 25% drug effect with 80% power at a 0.05 test level.

### Results

A total of 716 individuals were included in the analysis: 413 cognitively unimpaired (CU) participants (58.6% female; mean age = 70.6 years, SD = 7.9) and 303 cognitively impaired (CI) participants (54.7% female; mean age = 73.2 years, SD = 7.4). In A $\beta$ -positive individuals, the annual rate of change in plasma p-tau217 was similar between CU (0.07 pg/mL/y, SD = 0.11) and CI (0.08 pg/mL/y, SD = 0.13) groups. Effect size was 0.64 and 0.62 in CU and CI A $\beta$ -positive individuals, respectively. The minimum sample size required per study group to detect a 25% drug effect was 610 for the CU A $\beta$ -positive and 664 for the CI A $\beta$ -positive group. Notably, selecting individuals with intermediate A $\beta$  levels (Centiloid 20–40) yielded higher effect sizes (CU: 0.85; CI: 0.72), which reduced the required sample sizes per study group to 342 for CU and 492 for CI.

## MORE ONLINE

## Supplementary Material

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## Glossary

**AD** = Alzheimer disease; **ADNI** = Alzheimer's Disease Neuroimaging Initiative; **A $\beta$**  =  $\beta$ -amyloid; **BICWALZS** = Biobank Innovations for Chronic Cerebrovascular Disease With Alzheimer's Disease Study; **CI** = cognitively impaired; **CU** = cognitively unimpaired; **MCI** = mild cognitive impairment; **MMSE** = Mini-Mental State Examination; **MYHAT-NI** = Monongahela-Youghiogheny Healthy Aging Team-Neuroimaging; **NfL** = neurofilament light chain; **p-tau** = phosphorylated tau; **WRAP** = Wisconsin Registry for Alzheimer's Prevention.

## Discussion

Our findings support that changes in plasma p-tau217 represent a robust endpoint for clinical trials targeting CU or CI individuals with A $\beta$  pathology.

## Introduction

Biomarkers that reflect  $\beta$ -amyloid (A $\beta$ ) plaque accumulation and tau neurofibrillary tangles are key indicators of Alzheimer disease (AD) pathology.<sup>1</sup> AD brain pathology is typically assessed with PET or CSF tests, which are often used as endpoints in clinical trials for disease staging and progression monitoring.<sup>2-5</sup> However, these methods are constrained by invasiveness, cost, and access. As a result, interest in blood-based biomarkers has grown because they offer a more practical and scalable approach for both clinical diagnostics and research settings.

Studies using well-defined patient cohorts demonstrate the potential for blood biomarkers of core AD pathology—such as phosphorylated tau (p-tau) and A $\beta$ —as well as neurodegeneration markers such as neurofilament light chain (NfL).<sup>6-9</sup> These studies show that plasma biomarkers in individuals with AD exhibit abnormal levels and closely reflect CSF measures, underscoring their diagnostic utility.<sup>10</sup> This supports the use of blood biomarkers as valuable tools for early and accurate diagnosis, prognosis, and for tracking disease progression and treatment response in clinical settings.<sup>11</sup> Moreover, in symptomatic individuals receiving anti-A $\beta$  therapy, shifts in plasma biomarker levels have been used to assess treatment efficacy and disease modification.<sup>5,12-14</sup> Our group recently reported that the plasma biomarkers p-tau181 and NfL may serve as potential endpoints in clinical trials focusing on preclinical AD.<sup>15</sup> Taken together, these observations support the use of blood biomarkers to track drug effects in clinical trials.

Numerous studies have shown that plasma p-tau217 levels are associated with both A $\beta$  and tau pathologies.<sup>16-20</sup> Recent longitudinal findings have demonstrated that changes in plasma p-tau217 correlate with cognitive decline,<sup>19,21</sup> supporting its potential role as a biomarker for tracking disease progression. Moreover, findings from recent clinical trials suggest that plasma p-tau217 may be sensitive to treatment effects.<sup>12</sup> Collectively, these findings underscore the potential utility of plasma p-tau217 changes as a marker for monitoring AD progression in clinical trials. Although plasma p-tau217 has been used as an exploratory endpoint in recent trials,<sup>12,22</sup> key questions regarding its effect size, the sample size required

to detect meaningful changes—particularly in the context of its potential use as a primary endpoint—remain unexplored. This study assessed the utility of plasma p-tau217 as an endpoint in clinical trials targeting early AD.

## Methods

### Study Population and Clinical Assessment

This study included 716 individuals from 4 distinct cohorts, each with longitudinal plasma p-tau217 measurements: Alzheimer's Disease Neuroimaging Initiative (ADNI), the Biobank Innovations for Chronic Cerebrovascular Disease With Alzheimer's Disease Study (BICWALZS), the Monongahela-Youghiogheny Healthy Aging Team-Neuroimaging (MYHAT-NI) study, and Wisconsin Registry for Alzheimer's Prevention (WRAP). eFigure 1 provides a flowchart detailing inclusion numbers for each cohort. The ADNI<sup>23</sup> is a multicenter, observational cohort enrolling participants at more than 50 sites across the United States and Canada under the leadership of Michael W. Weiner, MD. The BICWALZS cohort comprises individuals seen in memory clinics at 5 university hospitals and 1 community geriatric mental-health center in South Korea. Participants were voluntarily recruited through referrals from neurology or psychiatry outpatient memory clinics. MYHAT-NI recruitment took place in a small community in southwestern Pennsylvania, part of the nation's historically industrial Rust Belt.<sup>24</sup> Participants were selected through age-stratified random sampling from voter registration records. They had to be 65 years or older and reside within the catchment areas. A neuroimaging substudy enrolled a subset of active participants who did not exhibit cognitive impairment. The WRAP study is a US observational cohort with longitudinal follow-up, enriched for participants reporting a parental history of probable AD dementia.<sup>25</sup> Cognitive status was defined based on a consensus panel review.<sup>25</sup> All participants provided written informed consent. Participants were categorized into 3 groups based on their global Clinical Dementia Rating (CDR) score: a score of 0 CU; a score of 0.5 corresponded to mild cognitive impairment (MCI); and scores above 0.5, in conjunction with

meeting the NIA-Alzheimer's Association criteria for probable Alzheimer disease, were classified as AD dementia.<sup>1</sup> For individuals without an available CDR score, the clinical diagnosis was used to determine group classification. For the purposes of this study, the MCI and AD dementia groups were classified as cognitively impaired (CI). Within the CI group, functional impairment severity was further stratified using CDR sum of boxes (CDR-SB) scores into 3 ranges: 0.5–2.5, 3.0–4.0, and 4.5–9.0.<sup>26</sup>

### Standard Protocol Approvals, Registrations, and Patient Consents

The ADNI study was conducted in accordance with ICH-GCP, 21 CFR 50/56, and HIPAA requirements (ClinicalTrials.gov: NCT02854033). Procedures for the BICWALZS study received approval from the Institutional Review Board of Asan Medical Center (IRB No. AJIRB-BMR-SUR-16-362; approval date: November 11, 2016). All protocols for the MYHAT-NI study were approved by the University of Pittsburgh Institutional Review Board (IRB: STUDY19020264; approval date: January 11, 2023). In the WRAP study, all protocols were approved by the Institutional Review Board at the University of Wisconsin-Madison (IRB: 2023-1522; approval date: January 23, 2024). Written informed consent was obtained from all participants before enrollment in each study.

### MRI/PET Biomarkers

Aβ-PET imaging was performed using [<sup>18</sup>F]Florbetapir (ADNI), [<sup>11</sup>C]PiB (MYHAT-NI and WRAP), and [<sup>18</sup>F]Flutemetamol (BICWALZS) radiotracers. For the ADNI, BICWALZS, and MYHAT-NI data sets, standardized uptake value ratios (SUVs) were converted into Centiloid units using a validated methodology.<sup>27</sup> In the WRAP cohort, Aβ-PET measures were determined using a previously published protocol to transform the distribution volume ratio (DVR) to the Centiloid scale.<sup>28,29</sup> Aβ-PET positivity for all cohorts was defined using Centiloid values equal to or greater than 20.<sup>16</sup> Participants with Centiloid values between 20 and 40 were classified as having intermediate Aβ levels.<sup>30,31</sup>

### Plasma p-tau Quantification

Plasma p-tau217 was quantified using the ALZpath assay on the single-molecule array (Simoa) platform.<sup>32</sup> This is a commercially available, ultra-sensitive plasma-based assay that uses a proprietary monoclonal antibody specific to p-tau217 for capture, a detection antibody targeting the N-terminus, and a peptide calibrator. Analyses within the ADNI cohort were conducted under the framework of the Foundation for the NIH Biomarkers Consortium initiative.<sup>33</sup> Plasma p-tau217 measurements for the BICWALZS and MYHAT-NI cohorts were performed at the University of Pittsburgh, Pittsburgh, PA. Plasma p-tau217 measurements in WRAP were conducted at the Clinical Neurochemistry Laboratory, University of Gothenburg, Gothenburg, Sweden.

### Statistical Analysis

The annual rate of change was calculated as follow-up minus baseline divided by the time interval:  $\left( \frac{\text{Follow up} - \text{Baseline}}{\Delta \text{time}} \right)$ .

Plasma p-tau217 values were normalized to z-scores within each cohort, using CU Aβ-negative participants as the reference group. Associations between plasma p-tau217 measures and Mini-Mental State Examination (MMSE) scores were examined using linear mixed-effects models with cohort included as a covariate, and model assumptions were verified before analysis. In addition, a separate regression model with dummy-coded Clinical Dementia Rating-Sum of Boxes (CDR-SB) groups was conducted to evaluate the association between changes in plasma p-tau217 levels and differences between groups. Cohort was included as a covariate in the models. Next, we applied LOWESS (locally weighted scatterplot smoothing)—a nonparametric method that fits a smooth curve to a scatterplot—using 1,000 iterations and a smoothing span of 0.95 to characterize annual changes in plasma p-tau217 across baseline CDR-SB scores.<sup>34</sup> We calculated effect size as the mean change in biomarker divided by its SD.<sup>3</sup> Using longitudinal plasma p-tau217 change as primary outcome, we then estimated the sample size needed to detect 25% reduction in biomarker levels with 80% power and a 5% significance level (eFigure 2). These calculations followed the validated formula and parameters from previous studies<sup>3,35,36</sup> (eMethods 1), to increase comparability with other studies using the same parameters.<sup>2,3,15</sup> To estimate the recruitment sample size for each group, we considered the prevalence of Aβ positivity in our study population. The cost of biomarkers and clinical assessments was calculated based on a previous study and is described in the supplemental methods (eMethods 2).<sup>15,37</sup> All statistical analyses were conducted using R statistical software version 4.0.5.

### Data Availability

Requests to access the raw or processed data and study materials by contacting the corresponding author (T.A.P.). Deidentified data will be provided to qualified academic investigators for the purpose of replicating the study's methods and findings. Information about the WRAP cohort is available at wrap.wisc.edu; ADNI data can be accessed at ida.loni.usc.edu. All sharable materials will be released under a material-transfer agreement, with some details withheld to protect participant privacy.

## Results

This study included a combined total of 716 participants from all cohorts. The mean follow-up durations of the CU and CI populations were 2.3 (SD = 0.6) and 2.0 (SD = 0.5) years, respectively. The CU group included 413 individuals from the ADNI, MYHAT-NI, and WRAP cohorts (58.6% female; mean age at baseline = 70.6 [SD = 7.9] years). The CI group included 303 individuals from the ADNI and BICWALZS cohorts (54.7% female; mean age at baseline = 73.2 [SD =

**Table 1** Demographics and Key Characteristics of the Study Population

|                                                | CU (N = 413) | CI (N = 303) |
|------------------------------------------------|--------------|--------------|
| Age at baseline, y, mean (SD)                  | 70.6 (7.9)   | 73.2 (7.4)   |
| Education, y, mean (SD)                        | 16.2 (2.8)   | 12.5 (5.4)   |
| Sex, female, n (%)                             | 242 (58.6%)  | 166 (54.7%)  |
| MMSE, mean (SD)                                | 28.9 (1.3)   | 25.9 (3.9)   |
| Race/ethnicity, n (%)                          |              |              |
| Asian                                          | 3 (0.7%)     | 138 (45.5%)  |
| Black                                          | 14 (3.4%)    | 4 (1.3%)     |
| White                                          | 390 (94.4%)  | 158 (52.2%)  |
| Other/unknown                                  | 6 (1.5%)     | 3 (1.0%)     |
| Aβ status at baseline, n (%)                   |              |              |
| Positive                                       | 109 (26.4%)  | 147 (48.5%)  |
| Intermediate Aβ levels (CL 20–40)              | 37 (8.9%)    | 38 (12.5%)   |
| Diagnosis, n (%)                               |              |              |
| CU                                             | 413 (100%)   | —            |
| MCI                                            | —            | 259 (85.5%)  |
| Dementia                                       | —            | 44 (14.5%)   |
| Plasma p-tau217 baseline (z-scored), mean (SD) | 0.72 (2.1)   | 1.77 (2.8)   |
| Follow-up time, y, mean (SD)                   | 2.3 (0.6)    | 2.0 (0.5)    |

Abbreviations: Aβ = β-amyloid; CI = cognitively impaired; CL = Centiloid; CU = cognitively unimpaired; MCI = mild cognitive impairment; MMSE = Mini-Mental State Examination; p-tau217 = phosphorylated tau 217.

7.4] years) comprising 259 (85.5%) MCI individuals. In the CU group, 109 individuals (26.4%) were Aβ-PET positive, whereas in the CI group, 147 individuals (48.5%) were Aβ-PET positive. In the CU group, 37 individuals (8.9%) had intermediate Aβ levels and 38 (12.5%) in the CI group. Demographics and key characteristics of the entire population and of individual cohorts are summarized in Table 1 and eTable 1.

### Changes in Plasma p-tau217 as a Function of Baseline Levels

Over 2 years, plasma p-tau217 levels increased significantly in both CU and CI (eFigures 3–5). In the CU group, baseline plasma p-tau217 was significantly associated with follow-up levels ( $\beta = 0.997$  95% confidence interval (CI) 0.94–1.03;  $p < 0.0001$ ; eTables 2 and 3, eFigure 6A). In the CI group, baseline plasma p-tau217 was also significantly associated with follow-up levels ( $\beta = 0.994$ ; 95% CI 0.95–1.02;  $p < 0.0001$ ; eTables 2 and 3, eFigure 6B). Results were consistent across cohorts (eTable 4).

### Association of Plasma p-tau217 With Cognitive Measures

Changes in plasma p-tau217 over time were significantly associated with the interaction between baseline MMSE and

time in the CI group ( $\beta = -0.02$ ; 95% CI  $-0.04$  to  $-0.005$ ;  $p = 0.01$ ), but not in the CU group ( $\beta = -0.001$ ; 95% CI  $-0.02$  to  $0.02$ ;  $p = 0.894$ ) Table 2. The model was internally validated

**Table 2** Predictors of Annual Changes in Plasma p-tau217 Levels by Cognitive Status

|                                   | $\beta$ (95% CI)        | T Value | p Value |
|-----------------------------------|-------------------------|---------|---------|
| <b>Cognitively unimpaired</b>     |                         |         |         |
| MMSE baseline                     | −0.04 (−0.15 to 0.06)   | −0.77   | 0.441   |
| MMSE baseline × time <sup>a</sup> | −0.001 (−0.02 to 0.02)  | −0.13   | 0.894   |
| Aβ pathology <sup>b</sup>         | 2.5 (2.1 to 2.8)        | 14.8    | <0.0001 |
| <b>Cognitively impaired</b>       |                         |         |         |
| MMSE baseline                     | −0.02 (−0.09 to 0.04)   | −0.67   | 0.499   |
| MMSE baseline × time <sup>a</sup> | −0.02 (−0.04 to −0.005) | −2.5    | 0.01    |
| Aβ pathology <sup>b</sup>         | 2.9 (2.4 to 3.4)        | 13.8    | <0.0001 |

Abbreviations: Aβ = β-amyloid; CI = cognitively impaired; CL = Centiloid; CU = cognitively unimpaired; MMSE = Mini-Mental State Examination; p-tau217 = phosphorylated tau 217.

<sup>a</sup> Time from first measure.

<sup>b</sup> Aβ status (positive and negative). The model was also corrected by cohort. Potential confounders were included in the models as covariates: age, sex and years of education.



using bootstrap analysis (eTable 5), and results were consistent across cohorts (eTable 6). Within the CI group, those with a baseline CDR-SB score > 4.5 exhibited greater annual increases in plasma p-tau217 compared with those with lower scores (eFigure 7).

### Longitudinal Changes and Effect Size on Plasma of p-tau217

We calculated the annual rate of change in plasma p-tau217 and its effect size for the CU and CI groups (Figure 1, eFigure 8). In both groups, the annual change in plasma p-tau217 was significantly different from zero in the whole population (CU mean: 0.03; 95% CI 0.025–0.040; CI mean: 0.04; 95% CI 0.024–0.049), among A $\beta$ -positive individuals (CU mean: 0.07; 95% CI 0.048–0.089; CI mean: 0.08; 95% CI 0.053–0.099) and those with intermediate A $\beta$  levels (CU mean: 0.06; 95% CI 0.038–0.088; CI mean: 0.05; 95% CI 0.020–0.072).

We evaluated sex differences in the annual change of plasma p-tau217 levels (eFigures 9 and 10). We found that in the CU (female mean = 0.03 [SD = 0.08]; male mean = 0.04 [SD = 0.08]) and CI (female mean = 0.04 [SD = 0.13]; male mean = 0.03 [SD = 0.08]) groups, there were no significant differences by sex.

In the whole population, regardless of A $\beta$  status, the effect size was 0.43 for the CU and 0.33 for the CI group (eTable 7). Within the A $\beta$ -positive population, the effect size was 0.64 for the CU and 0.62 for the CI group (eTables 7 and 8). In individuals with intermediate A $\beta$  levels, the effect size was 0.85 for the CU and 0.72 for the CI group.

### Sample Size Required for Clinical Trials

To estimate the number of participants needed for a hypothetical clinical trial assessing a 25% longitudinal reduction in

plasma p-tau217, we calculated the required sample size for clinical trials targeting CU and CI populations (Figure 2A, eTables 7 and 8). We found that a clinical trial enrolling CU individuals who are A $\beta$ -positive would require 610 individuals per study group, while for a clinical trial enrolling only CI A $\beta$ -positive individuals, the sample size required would be 664 individuals per study group. When enrolling only individuals with intermediate levels of A $\beta$ , the required sample size per study group would be 342 for the CU group and 492 for the CI group.

### Cost Analysis of Plasma p-tau217 as Endpoint for Clinical Trials

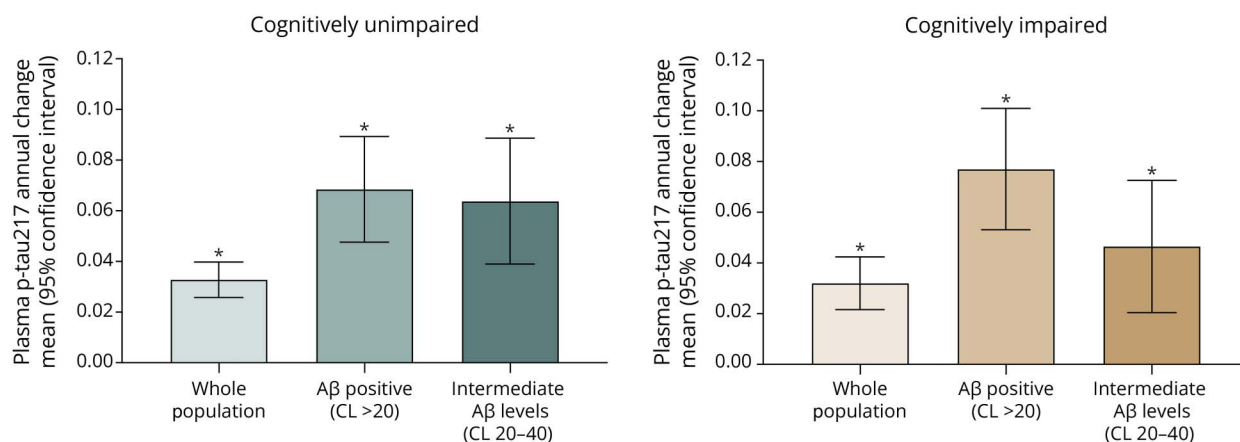
Figure 2A illustrates the estimated cost of a clinical trial, including costs related to participant recruitment, population enrichment, clinical assessments, and the use of plasma p-tau217 as a primary endpoint. Among CU A $\beta$ -positive individuals, the total trial cost using plasma p-tau217 was estimated at \$22.4 million. Notably, by selecting individuals with intermediate A $\beta$  levels, the total cost reduced by 54% (to \$10.3 million).

In the CI A $\beta$ -positive group, the estimated cost was \$13.8 million, which reduced by 26% (to \$10.2 million) when including only individuals with intermediate A $\beta$  levels.

### Sample Size Estimation as a Function of Multiple Drug Effects and Power Levels

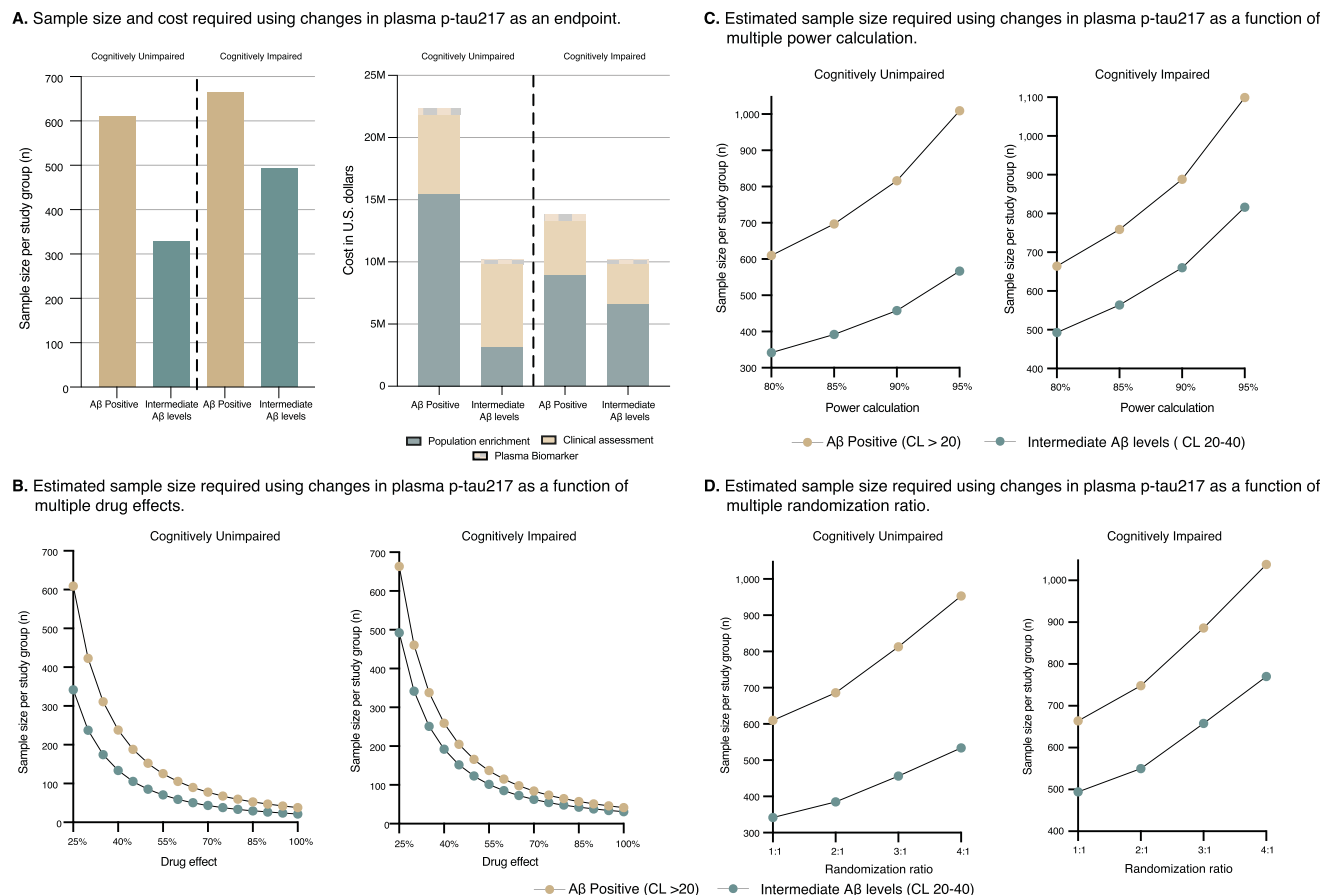
We estimated the sample size requirements for detecting different drug effects on the reduction of plasma p-tau217. As shown in Figure 2B, required sample sizes decrease as drug effects increase. Next, we examined how sample size estimates vary across different statistical power thresholds (80%, 85%, 90%, and 95%) for trials using longitudinal changes in plasma p-tau217 as the endpoint. As shown in Figure 2C, there is a progressive increase in the required sample size per study

**Figure 1** Annual Change in Plasma p-tau217 Levels Across Clinical Groups



The plot shows the annual change in plasma p-tau217 levels (pg/mL/y) along with the corresponding 95% CIs for the CU and CI population. \*Indicates that the 95% CI does not cross zero, meaning a statistically significant change. A $\beta$  =  $\beta$ -amyloid; CI = cognitively impaired; CL = Centiloid; CU = cognitively unimpaired; p-tau217 = phosphorylated tau 217.

**Figure 2** Plasma p-tau217 as an Endpoint in AD Clinical Trials



Plot A shows the sample size per study group and estimated trial cost using changes in plasma p-tau217 to monitor drug effects in CU and CI individuals who are A $\beta$ -positive and have intermediate A $\beta$  levels. The plot B shows the estimated sample size per study group as a function of multiple hypothesized drug effects (greater than the tested 25% reduction in the rate of biomarker progression) in the CU and CI groups. The plot C shows the estimated sample size per study group as a function of varying statistical power levels in the CU and CI groups. The plot D shows the estimated sample size per study group as a function of different randomization ratios in the CU and CI groups. For all calculations, the following cost assumptions were used: PET = \$3,000; plasma biomarker = \$200; recruitment/consenting/clinical assessment = \$1,000. Costs for assessments and plasma biomarkers were calculated for 2 time points (baseline and follow-up). Biomarker and procedure costs were based on estimates from U.S. research settings. An attrition rate of 10% was applied.  $\Delta$  = annual rate of change. Additional cost details are provided in eMethods 2. A $\beta$  =  $\beta$ -amyloid; CI = cognitively impaired; CL = Centiloid; CU = cognitively unimpaired; p-tau217 = phosphorylated tau 217.

group with higher power levels, observed consistently in both CU and CI individuals who are either A $\beta$ -positive or exhibit intermediate A $\beta$  levels. Figure 2D illustrates how sample size requirements are further influenced by varying allocation ratios (e.g., 2:1, 3:1, and 4:1). We also conducted a range-based sensitivity analysis of sample size by varying the mean (and SD) change. eFigure 11 illustrates how the required sample sizes vary across different trial scenarios.

## Discussion

In this study, we found that individuals with higher baseline plasma p-tau217 levels exhibited more pronounced longitudinal increases in biomarker levels, which were also linked to cognitive decline. Our cost analysis indicated that trials enrolling CU or CI participants with intermediate A $\beta$  levels would result in lower overall costs compared with trials using

A $\beta$  positivity alone as an enrichment strategy. Our findings provide a blueprint for the potential use of plasma p-tau217 as an endpoint in early AD clinical trials.

We found that individuals with greater baseline AD pathophysiology exhibited steeper longitudinal increases in plasma p-tau217 concentrations. These changes were not significantly different when stratified by sex. Specifically, individuals with elevated baseline plasma p-tau217 levels exhibited the most pronounced rate of increase over time, aligning with trends reported in previous studies.<sup>38</sup> Similarly, individuals with worse baseline cognitive test scores exhibited more pronounced changes in plasma p-tau217 over time, indicating that individuals with greater cognitive impairment are more likely to exhibit substantial changes in plasma p-tau217 levels. The more modest change in plasma p-tau217 among individuals with early pathology suggests that the development of other blood tests/assays may be needed for clinical trials

designed to track AD progression in its earliest stages. In contrast to recent tau PET studies that have identified sex-related differences in longitudinal changes,<sup>39</sup> our analysis did not reveal significant difference between men and women in plasma p-tau217 trajectories over time. Our findings align with a previous report<sup>40</sup> indicating no significant sex differences in longitudinal plasma p-tau217 levels. This suggests differences in the underlying pathophysiology of tau-PET and plasma p-tau.

Our findings suggest that longitudinal changes in plasma p-tau217 can effectively monitor large phase 3 clinical trials. However, this biomarker may lack sufficient power to detect drug effects in smaller phase 1–2 trials. Our approach is consistent with recent clinical trials, such as TRAILBLAZER-ALZ, which used plasma p-tau217 as an exploratory endpoint and reported reductions in biomarker levels among participants receiving treatment.<sup>12</sup> Moreover, consistent with previous studies,<sup>15</sup> we demonstrated that population enrichment strategies—such as including A $\beta$ -positive individuals or those with intermediate A $\beta$  levels—lead to higher effect sizes and, consequently, reduced sample sizes. Notably, a previous study from our group demonstrated that plasma p-tau181 could serve as an outcome biomarker in a large-scale population trial (>2,000 participants).<sup>15</sup> In this study, we build on those findings by showing that changes in plasma p-tau217 may allow for a substantially smaller sample size per study group (~600), further supporting its utility as an endpoint in adequately powered phase 3 trials. Furthermore, we found that changes in plasma p-tau217 are greater in individuals with more advanced cognitive impairment, in line with previous studies,<sup>19,38</sup> and supporting its potential for monitoring cognitive decline in later disease stages. Together, our results provide a blueprint for the formal use of plasma p-tau217 as an endpoint in AD clinical trials.

Our results demonstrate that plasma p-tau217 may serve as a potential endpoint for optimizing the design and cost-efficiency of AD clinical trials. In this study, we assessed the effect of different population enrichment strategies on sample size requirements for clinical trials. Notably, selecting individuals with intermediate levels of A $\beta$  burden reduced the required sample size—particularly among CU participants, who exhibited lower interindividual variability. This resulted in larger effect sizes, which in turn led to reduced sample size estimates. These results were consistent across varying levels of statistical power, assumed treatment effects, and randomization ratios, further supporting previous evidence that selecting individuals with intermediate A $\beta$  levels may enhance the efficiency of clinical trial designs.<sup>15</sup> Finally, our exploratory cost analysis demonstrated that applying enrichment strategies can lead to direct cost savings by reducing the required sample size for clinical trials. Specifically, the subgroup enriched for intermediate A $\beta$  levels demonstrated a cost reduction of 54% among CU participants and 26% among CI participants. Taken together, these findings underscore the potential of intermediate A $\beta$  levels as a strategic criterion for

enriching clinical trials that use changes in plasma p-tau217 as an endpoint.

One of the strengths of this study lies in its integration of data from both specialized memory clinics and community-based cohorts representing multiple ethnic backgrounds, which contributes to broader applicability. However, these results should be considered in the context of certain limitations. Despite the inclusion of participants from White and Asian backgrounds, the representation of Black and Hispanic individuals was notably limited. Replicating these results in larger, more demographically varied population cohorts would bolster confidence in their broader applicability.<sup>41,42</sup> Although a 25% drug effect is a commonly used assumption, changing this parameter could alter our sample size requirements and conclusions. Future studies that evaluate plasma p-tau217 as a clinical trial endpoint should incorporate formal non-inferiority or superiority comparisons with established biomarkers to better contextualize its clinical relevance and utility. The number of individuals with intermediate A $\beta$  levels was low in each cohort, so we were only able to perform the analysis on the combined population rather than within each cohort separately. Finally, the use of longer and more frequent follow-up plasma p-tau217 collections could yield more robust longitudinal estimates and larger effect sizes.<sup>19,38</sup>

In conclusion, our findings suggest that plasma p-tau217 is a promising marker for tracking treatment response in both preclinical and symptomatic A $\beta$ -positive individuals.

### Author Contributions

P.C.L. Ferreira: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. B. Bellaver: drafting/revision of the manuscript for content, including medical writing for content. G. Povala: drafting/revision of the manuscript for content, including medical writing for content. G. Bauer-Negrini: analysis or interpretation of data. C.S. Aguzzoli: analysis or interpretation of data. J. P. Ferrari-Souza: drafting/revision of the manuscript for content, including medical writing for content. D.T. Leffa: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. C. Soares: analysis or interpretation of data. F.Z. Lussier: drafting/revision of the manuscript for content, including medical writing for content. M.S. Medeiros: drafting/revision of the manuscript for content, including medical writing for content. C. Felix: drafting/revision of the manuscript for content, including medical writing for content. E. Ruppert: drafting/revision of the manuscript for content, including medical writing for content. F. Rohden: analysis or interpretation of data. W.V. Borelli: analysis or interpretation of data. H. Karim: major role in the acquisition of data. R. Langhough: major role in the acquisition of data. B.T. Christian: major role in the acquisition of data. R. Wilson: major role in the acquisition of data. C.H. Hong: major role in the acquisition of data. H.W. Roh: major

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WO2020193500A1: Use of a ps396 assay to diagnose tauopathies; US 63/679,361: Methods to Evaluate Early-Stage Pre-Tangle TAU Aggregates and Treatment of Alzheimer's Disease Patients; US 63/672,952: Method for the Quantification of Plasma Amyloid-Beta Biomarkers in Alzheimer's Disease; US 63/693,956: Anti-tau Protein Antigen Binding Reagents; and 2450702-2: Detection of oligomeric tau and soluble tau aggregates. B. Snitz, S. Johnson, S.J. Son, and T.A. Pascoal report no disclosures relevant to the manuscript. Go to [Neurology.org/N](https://www.neurology.org/N) for full disclosures.

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